

Sebastien Fernandez

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Summary

Mechanical Engineering graduate (Dec 2025) with hands-on experience in mechanical design, prototyping, simulation, and electromechanical systems integration. Skilled in SolidWorks modeling, MATLAB-based analysis, and physical system testing. Seeking entry-level roles in mechanical design, manufacturing, test, or product development.

Education

California State University Long Beach (CSULB)

Bachelor of Science in Mechanical Engineering

December 2025

Relevant Coursework: Machine Design, Thermodynamics, Fluid Mechanics, Dynamics, Control Systems, Materials Engineering, Manufacturing Processes

Technical Skills

Mechanical Design & CAD

SolidWorks (part modelling, assemblies, drawings), Fusion 360, AutoCAD

Analysis & Simulation

MATLAB (dynamic system modeling, numerical simulation, data visualization)

Manufacturing & Prototyping

FDM 3D printing, manual lathe operation, mechanical assembly, soldering

Programming & Data Tools

C++, Python, SQL, VBA, Microsoft Access

Projects

Remote Operated Submersible — Project Leader

- Led multidisciplinary team responsible for propulsion, ballast, structural design, sealing, and electronics integration
- Designed adjustable ballast system to achieve neutral buoyancy and improve pitch stability
- Evaluated mass distribution and center-of-buoyancy alignment to enhance underwater stability
- Coordinated system assembly and resolved mechanical integration issues during testing

Mechanical System Design — Multi-Joint Quadruped Platform

- Designed and modeled 25+ mechanical components in SolidWorks for a multi-joint load-bearing structure
- Optimized geometry for structural integrity, weight distribution, and part interchangeability
- Fabricated and assembled full prototype using FDM 3D printing
- Integrated mechanical housing with electronic components to ensure stability and serviceability

Projectile Motion Simulation — MATLAB

- Developed numerical simulation modeling 2D projectile motion including velocity, displacement, and energy profiles
- Parameterized model to adapt to varying mass and initial conditions
- Generated graphical outputs for performance evaluation and validation against analytical solutions

Relational Database System — Simulated Business Application

- Designed relational database structure with linked tables to maintain data integrity
- Implemented SQL queries for record insertion, renewal tracking, and data retrieval
- Structured tier-based membership model with automated data handling logic